

The arithmetic of the kernel

How RVD's predicates get exact signs: error-free transformations, expansions,
the filter, and Simulation of Simplicity

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What this is. A reference to come back to when the arithmetic of `code/src/kernel/` stops making sense. It explains, from the bottom up, why a predicate cannot simply be evaluated in `double`, and how each layer of the kernel fixes one part of that: the error-free transformations of `eft.h`, the expansions of `expansion.h` and `dynamic_expansion.h`, the filter of `bounded.h`, and the symbolic perturbation that step 7 adds on top. Each section gives the theory, says where it lives in the code, and gives a worked example small enough to check by hand. The companion document `documents/predicates/side_predicates.tex` derives the formulas the predicates evaluate; this one is about *how* they are evaluated.

State, 5 October 2026. Everything up to `side4` is written and tested. Symbolic perturbation (§8) is designed and its tests are written and validated; `side1_sos` is written and passes, `side2_sos` to `side4_sos` are next. The measured numbers are in §9, with how to reproduce them.

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1 The problem, and the plan

1.1 A predicate is the sign of a polynomial

Every decision the clipper takes is a *predicate*: a question with a discrete answer, such as “is this vertex on seed i ’s side of the bisector of i and m ?”. Each one is the sign of a polynomial in the input numbers: `orient2d` is a 2×2 determinant of coordinate differences (degree 2), `orient3d` a 3×3 one (degree 3), and the side predicates are ratios of determinants of degree up to 6 over 4. The value itself is never needed, only whether it is negative, zero or positive.

1.2 Why doubles get the sign wrong

A `double` holds 53 significant bits, and each operation rounds its exact result to the nearest double. A polynomial of degree 6 in 53-bit inputs can need a few hundred bits to be written exactly, and near a degeneracy – a point almost on a line, a vertex almost on a bisector – the exact value is tiny while the rounding errors of the terms that nearly cancel are not. Then the computed value can have the wrong sign, or be zero when the true value is not, or the other way round.

This is not a rare corner case in a Voronoi code. Two cells look at the same vertex, each through its own computation, and if their roundings disagree, one cell thinks a facet exists and the other does not: the diagram has a crack. On a regular lattice of seeds, every vertex

is a degeneracy. Measured on the side predicates' test inputs (§9): on ties moved by one ulp, constructing the vertex in doubles and comparing distances got 65% to 76% of the signs wrong.

1.3 Three layers

The kernel computes each sign in up to three steps (figure 1), following DESIGN.tex, section 6.3:

1. **The filter** (§6). Evaluate the polynomial in doubles, and alongside it a certified bound on the rounding error. If the value is farther from zero than the bound, its sign is proven. On random inputs this decided 98% to 100% of calls, at 3 to 13 times the cost of a plain double evaluation (§9).
2. **The exact path** (§3 to §5). Otherwise evaluate the same polynomial exactly, as an *expansion*: an unevaluated sum of doubles built from error-free operations. Its sign is the true sign, zero included.
3. **Symbolic perturbation** (§8). If the exact value is zero, answer the question for an imaginary, infinitesimally perturbed input instead, the same one for every predicate, so that ties are broken consistently and no predicate ever answers 0.

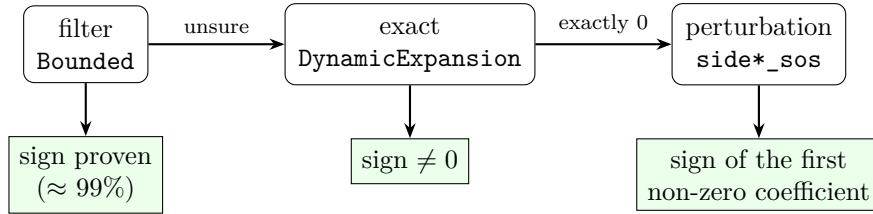


Figure 1: How one predicate call is decided. Each step runs only if the one before could not decide. The counters of `predicate_counts.h` record which step did.

The rest of this document goes bottom-up: floating point (§2), the error-free transformations (§3), expansions (§4), the two containers (§5), the filter (§6), how a predicate combines them (§7), and the perturbation (§8).

2 Floating point, in one page

2.1 What a double is

So binary64 number has the components: sign, significand, and exponent. In binary notation $x = (-1)^s(1.f_1f_2 \dots f_{52})_2 2^{E-1023}$. So sign takes 1 bit, exponent takes 11 bits, and the significand takes then 52 bits. But since every normalized binary number, like shown in the binary notation, has the first bit 1, we can ignore that and "win" one more bit for the significand. This win in precision is called **implicit 1** or **hidden bit**. This does not mean 53 bits are stored but just assumed in the normal case that the leading bit is 1 and it is the most significant bit. The exponent is stored biased. Meaning $e = E - 1023$, so that we do not have to deal with signed bit ordering, it is just unsigned bit ordering for 11 bits. And like this we also can reserve all-zero or all-one bit patterns for special values. On the computer the bits are stored in the order: sign(1 bit), exponent (11 bits), significand (52 bits).

Two quantities appear everywhere:

- $\text{ulp}(x)$, the *unit in the last place*: the gap between x and the next double away from zero. For $1 \leq |x| < 2$ it is 2^{-52} .
- $u = 2^{-53}$, the *unit roundoff*: half an ulp of 1. In `bounded.h` it is `kU`.

For a **normal** number we have $E \in 1, \dots, 2046$. The exponent values $E = 0$ (zero or subnormal) and $E = 2047$ (∞ or *NaN*) are deliberately reserved.

When $E = 0$, the hidden 1 is removed and the formula becomes

$$x = (-1)^s (0.f)_2 2^{-1022}$$

these are subnormal numbers. Why -1022 ? With the smallest normal $E = 1$, we get the smallest normal exponent $e = -1022$. So the smallest normal positive number is

$$1.0 \dots 0_2 \times 2^{-1022} = 2^{-1022}$$

Now we also want to represent numbers below this, otherwise we jump directly to 0 after this smallest normal number. **Subnormals** fill that region. With this, the smallest positive binary64 number is 2^{-1074} .

Another special case is when $E = 0$ and all significand bits 0. Then $x = (-1)^s 0.0 \dots 0 2^0$. This gives two floating point zeros, one with positive sign and the other with a negative sign.

Another special case is when $E = 2^{12} - 1$. Here the significand bits determine if we get infinity or *NaN*. If significand bits are zero we get infinity (again with two signs). When significand bits are not zero then we get "not a number".

2.2 Rounding

Every operation $\circ \in \{+, -, \times\}$ returns $\text{fl}(a \circ b)$, the exact result rounded to the nearest double; a tie, exactly halfway between two doubles, goes to the one whose last bit is 0 (*round to nearest, ties to even*, the IEEE 754 default). In the normal range,

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| \leq u.$$

That one inequality is all the filter (§6) needs. The exact path needs three sharper facts (listed in `eft.h` as F1–F3):

F1 The rounding error of a sum is a double: $a + b - \text{fl}(a + b)$ is representable, and at most $\frac{1}{2} \text{ulp}(\text{fl}(a + b))$.

F2 *Sterbenz's lemma*: if $b/2 \leq a \leq 2b$, then $a - b$ is computed exactly.

F3 The rounding error of a product is a double, provided it does not underflow: the exact product of two 53-bit significands has at most 106 bits, so $ab - \text{fl}(ab)$ has at most 53.

2.3 What the code must not let the compiler do

These facts hold for the arithmetic they were proved for, so the build protects them:

- **No `-ffast-math`**: it lets the compiler use real-number algebra, under which the error terms below are identically zero and get deleted.
- **`-ffp-contract=off`**, for every target: it stops the compiler fusing $a \times b + c$ into one `fma`, which rounds once instead of twice and breaks the proofs. An explicit `std::fma`, as in `two_product`, is still honoured.
- **No extended precision**: x86-64 and ARM compute doubles in 64-bit registers, so the old x87 “double rounding” problem does not arise.

2.4 The range

The guarantees fail near the ends of the range, and what fails is the *error-free transformation*, not just the rounded operation:

- `two_sum` and `fast_two_sum` are exact unless the sum overflows. With gradual underflow the rounding error of a sum is always representable, even among subnormals.
- `two_product` is exact when the residual $ab - \text{fl}(ab)$ is exactly representable. That can fail near underflow: a residual below the subnormal grid is lost. A simple *sufficient* condition, the one the code relies on, is $|ab| \geq 2^{-969}$ ($= 2^{-1022+53}$) or $ab = 0$.

The condition is on *every intermediate product*, not on the inputs: two normal inputs near 2^{-600} already have a product that underflows. A predicate of degree 6 multiplies six inputs, so inputs must be zero or of moderate size: the side-predicate tests keep exponents in $[-20, 20]$, and a first draft that nudged a zero coordinate to the smallest subnormal (2^{-1074} , whose square underflows to 0) broke exactness at once. Checking the range belongs where a domain or a seed set is built.

3 Error-free transformations

An *error-free transformation* (EFT) turns one rounded operation into two doubles whose *sum is the exact result*:

$$\text{two_sum}(a, b) = \{h, \ell\} : \quad h = \text{fl}(a + b), \quad h + \ell = a + b \text{ exactly};$$

and the same for products. The second double ℓ is the rounding error, which F1 and F3 say is itself a double; the algorithms compute it exactly, using nothing but more rounded operations. The results of the sum and product EFTs (`two_sum`, `fast_two_sum`, `two_product`, `two_product_dekker`) also satisfy $|\ell| \leq \frac{1}{2} \text{ulp}(h)$: the two parts cover disjoint ranges of bits, and $h + \ell$ rounds back to h . `split` is different (§3.4): its parts do not overlap either, but its low part is only bounded by half a unit of the high part's 26th bit, which can be about 2^{27} times $\frac{1}{2} \text{ulp}(h)$. All of `kernel/eft.h` is five such functions, and everything exact in RVD is built from them.

3.1 fast_two_sum: three operations, one precondition

Shewchuk's Theorem 6 (after Dekker): if $|a| \geq |b|$, then

$$h = \text{fl}(a + b), \quad b_v = h - a, \quad \ell = b - b_v$$

gives $h + \ell = a + b$ exactly. The idea: because b is the smaller term, $h - a$ is computed exactly, so b_v is *exactly* the part of b that made it into h ; then $b - b_v$, the part that did not, is the rounding error, a double by F1, so that subtraction is exact too. Without $|a| \geq |b|$ the first subtraction can round and ℓ is wrong, which is why the code asserts the precondition, and why the tests compile with asserts on.

3.2 two_sum: six operations, no precondition

Shewchuk's Theorem 7 (after Knuth) removes the precondition by recovering both operands:

$$\begin{array}{lll} h = \text{fl}(a + b), & b_v = h - a, & a_v = h - b_v, \\ b_r = b - b_v, & a_r = a - a_v, & \ell = a_r + b_r. \end{array}$$

When $|b| > |a|$ the recovered b_v and a_v may themselves be rounded, but the errors made in recovering them cancel exactly in $a_r + b_r$.

Example. `two_sum`($2^{53}, 1$): the exact sum $2^{53} + 1$ lies halfway between the doubles 2^{53} and $2^{53} + 2$; the tie goes to the even one, so $h = 2^{53}$, and $\ell = 1$. The pair $\{2^{53}, 1\}$ holds the exact sum, which no single double can.

3.3 `two_product`: two operations, with fma

$h = \text{fl}(ab)$ and $\ell = \text{fma}(a, b, -h)$. The fma computes $ab - h$ with a single rounding; that exact value is a double by F3, so rounding it changes nothing, and ℓ is the exact error. It is fast wherever the CPU has an fma instruction: most x86-64 processors since Intel's Haswell and AMD's Piledriver, and every 64-bit ARM. Not all: Pentium- and Celeron-branded parts of those generations, and several Atom cores, have none, and then `std::fma` is emulated in software, slowly.

3.4 `split` and `two_product_dekker`: the same without fma

Shewchuk's Theorems 17 and 18 (Dekker, after Veltkamp). `split` cuts a into two halves of at most 26 significant bits each, using the constant $2^{27} + 1$; two such halves multiply to at most 52 bits, which fits a double exactly. So ab is the exact sum of four exact products of halves, and peeling them off h largest first, each subtraction exact, leaves the error. Kept because it is what to use where fma is emulated in software, and tested to give the same h and ℓ as `two_product` (compared with `==`, so a $+0$ and a -0 count as equal).

4 Expansions

4.1 Definition

An *expansion* is a finite list of doubles whose *exact* sum is the number it represents:

$$x = c_0 + c_1 + \cdots + c_{n-1} \quad \text{in the real numbers.}$$

Its components are kept in a strict normal form, the **invariant** that every operation assumes of its inputs and must restore in its result:

- **nonzero** – zero is the empty list, never a component;
- **nonoverlapping** – see below;
- **increasing magnitude** – smallest first, $|c_0| < |c_1| < \cdots < |c_{n-1}|$.

In both containers `append()` is the only way a component gets in. It drops zeros, asserts that each new component is larger in magnitude than the last, and in `Expansion<N>` asserts that there is room. It does *not* check nonoverlap: that is each operation's theorem, and the tests check it against exact rationals after every operation.

4.2 Nonoverlapping, nonadjacent, strongly nonoverlapping

Write a double's binary magnitude as a string of bits; say it *covers* the positions from its lowest set bit to its highest. Shewchuk's definitions (nonoverlapping, adjacent and nonadjacent in his section 2.1; strongly nonoverlapping in section 2.4, next to Theorem 13):

Definition 1. Two doubles x, y are **nonoverlapping** if the lowest set bit of one is above the highest set bit of the other: they cover disjoint ranges of positions. They are **adjacent** if they overlap, or if x overlaps $2y$, or $2x$ overlaps y : they touch, one's highest bit sitting right below the other's lowest. An expansion is

- *nonoverlapping* if no two components overlap;
- *nonadjacent* if no two components are adjacent;
- *strongly nonoverlapping* if it is nonoverlapping, no component is adjacent to two others, and any two adjacent components are both powers of two.

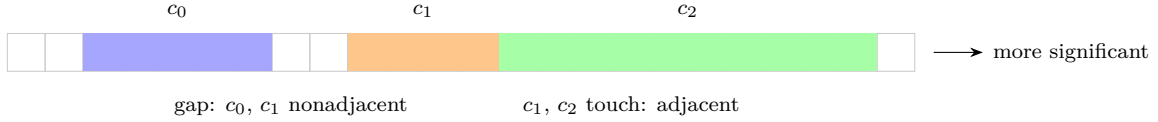


Figure 2: The bit ranges of a nonoverlapping expansion of three components, least significant on the left. c_1 and c_2 touch, so the expansion is nonoverlapping but not nonadjacent; and since they are not powers of two, not strongly nonoverlapping either.

4.3 What the invariant buys

Lemma 1 (the sign is the sign of the largest component). *In a nonoverlapping expansion, $|c_0 + \dots + c_{n-2}| < |c_{n-1}|$.*

Why. All the smaller components lie in bit positions below the lowest set bit of c_{n-1} , and in disjoint ranges, so their sum is less in magnitude than that lowest bit, which is at most $|c_{n-1}|$. $\square$

So a predicate’s answer is one comparison: the sign of the last component, or 0 for the empty expansion. This is `sign()`. But the largest component is *not* a good approximation of the value: $[1023.9, 1024]$ is nonoverlapping (all of 1023.9 lies below 1024’s single bit) and is worth about twice its largest component. To read a value, add the components smallest first: `approximate()`. A *compressed* expansion (§4.6) does have its largest component within an ulp of the value.

4.4 The operations

Each operation is exact: it returns the exact sum, product or negation of its operands, as another expansion. Theorem numbers are Shewchuk’s (1997).

grow(e, b) $e+b$ for a double b (Grow-Expansion, Theorem 10). Carry b up through the components with `two_sum`; each step’s error is final, because everything still to come is larger and does not overlap it; the last carry is the largest component.

sum(a, b) grow a by each component of b (Expansion-Sum, Theorem 12). $O(mn)$; kept as a reference for the faster ones.

fast_sum(a, b) merge a and b into one list by magnitude (as in merge sort), then one carry pass: `fast_two_sum` for the first two, `two_sum` after (Fast-Expansion-Sum, Theorem 13). $O(m + n)$. Requires *strongly* nonoverlapping inputs, and its promise to return one is false (§4.7); it is no longer used by the operators.

linear_sum(a, b) the same merge, with a carry kept as *two* doubles, Q and q (Linear-Expansion-Sum, Theorem 24, Shewchuk’s appendix): each new g_k first absorbs q with a `fast_two_sum`, whose low part is final, and the rest goes into Q with a `two_sum`. Nonoverlapping inputs give a nonoverlapping result, under any tie rule. About half again `fast_sum`’s work. **This is the sum the operators use.**

scale(e, b) $e \cdot b$ for a double b (Scale-Expansion, Theorem 19). Each component’s `two_product` is folded into the carry with a `two_sum` and a `fast_two_sum`: two final components per input component. Nonoverlapping in, nonoverlapping out; nonadjacent in, nonadjacent out.

product(a, b) $\sum_j \text{scale}(a, b_j)$, accumulated with `linear_sum`. Every step is a theorem.

negate(e) flip every sign; the bit ranges do not change.

compress(e) fewer components, same value (§4.6).

4.5 linear_sum step by step

With the merged list $g_1, \dots, g_{m+n}$ (by increasing magnitude), in the paper’s indexing:

$$\begin{aligned}
 (Q_2, q_2) &\leftarrow \text{fast_two_sum}(g_2, g_1) \\
 \text{for } i = 3 \dots m+n : \quad (R_i, h_{i-2}) &\leftarrow \text{fast_two_sum}(g_i, q_{i-1}) \quad h_{i-2} \text{ is final} \\
 (Q_i, q_i) &\leftarrow \text{two_sum}(Q_{i-1}, R_i) \\
 h_{m+n-1} &\leftarrow q_{m+n}, \quad h_{m+n} \leftarrow Q_{m+n}.
 \end{aligned}$$

Holding the error q back for one step, and letting the next component “clip a high-order bit off” it, is what makes the result nonoverlapping for any nonoverlapping inputs. Both **fast_two_sum** preconditions hold: the merge puts g_2 above g_1 , and q is the error of a rounding, at most half an ulp of Q , which the proof keeps close to g_i .

4.6 Compress

Products make expansions long: a product of an m - and an n -component expansion can have up to $2mn$ components, many of them a few bits that a neighbour could absorb. Shewchuk’s Compress (section 2.7, Theorem 23) runs two passes:

1. *downward*, largest to smallest: a running Q absorbs each component with **fast_two_sum** as long as the error is zero; at a nonzero error, Q is stored as finished and the error carries on;
2. *upward* over what pass 1 stored: the same carry as **grow**, merging what pass 1 left adjacent.

It returns a *nonadjacent* expansion, never longer than its input, whose largest component is within an ulp of the whole value. It does not promise the shortest form, and compressing again can change the length.

Example. $[1023.5, 1024]$ is a valid expansion whose largest component is half its value, but the value, 2047.5 , is a double; compressed, it is that one double.

4.7 Theorem 13 is false as stated

Fast-Expansion-Sum’s Theorem 13 promises: for strongly nonoverlapping inputs, on a round-to-even machine, a strongly nonoverlapping result. The result is nonoverlapping (Lemma 16 holds), but not always strongly nonoverlapping. The smallest binary64 example found:

$$[-1, -4, -2^{53}] + [2^{53} + 10] :$$

1. merged: $-1, -4, -2^{53}, 2^{53} + 10$;
2. **fast_two_sum** $(-4, -1) = \{-5, 0\}$;
3. **two_sum** $(-5, -2^{53})$: the exact $-2^{53} - 5$ lies halfway between $-2^{53} - 4$ and $-2^{53} - 6$ (doubles there are 2 apart); the tie goes to the even significand, $-2^{53} - 4$, and the error is -1 , output as a component;
4. **two_sum** $(-2^{53} - 4, 2^{53} + 10) = \{6, 0\}$, exactly.

The result is $[-1, 6]$: 6 covers bits 1–2 and $2 \cdot (-1)$ covers bit 1, so they are adjacent, and 6 is not a power of two. Both inputs are nonadjacent, so every hypothesis holds. The gap in the proof is Shewchuk’s footnote 5 (“intervening zeros ... Trust me”): an exact step with a zero error, sitting between two output components, lets a cancellation move low bits up into a later component. The details, and a 53-bit example found by the tests, are in the counterexample write-up and in `test_touching_inputs` of `tests/test_dynamic_expansion.cpp`.

Consequence for the code. A `fast_sum` result is a valid expansion, but not a *known-valid input* to another `fast_sum`, and every product and every $a + b + c$ is a chain of sums. So the kernel's only invariant is plain nonoverlap, the operators sum with `linear_sum` (Theorem 24 needs only nonoverlap), and every step of every exact evaluation is covered by a theorem.

5 Two containers: `Expansion<N>` and `DynamicExpansion`

The algorithms above say nothing about where the components are stored. The kernel has two answers, with the same operations and the same invariant.

5.1 `Expansion<N>`: the capacity in the type

`Expansion<N>` (`expansion.h`) holds at most N components in a `std::array` on the stack: no allocation, ever. Each operation's result type carries the worst-case length, so the compiler works out every size:

operation	result type
<code>grow(Expansion<N>, double)</code>	<code>Expansion<N + 1></code>
<code>linear_sum(Expansion<M>, Expansion<N>)</code>	<code>Expansion<M + N></code>
<code>scale(Expansion<N>, double)</code>	<code>Expansion<2N></code>
<code>product(Expansion<M>, Expansion<N>)</code>	<code>Expansion<2MN></code>
<code>negate(Expansion<N>)</code>	<code>Expansion<N></code>

A formula's inputs become `Expansion<1>`, and the type of the result falls out: `orient2d`'s determinant is an `Expansion<16>` and `orient3d`'s an `Expansion<192>`, both checked by a `static_assert`. As long as only the operations above are used, an expression's result type is large enough by construction. Two places still rely on runtime checks: `append()` asserts that there is room, and `fit<K>(e)`, the one way down to a smaller type, asserts that the components fit; it is used where a smaller bound is a theorem the types cannot see (inside `product`).

5.2 Why it stops working for the side predicates

The bound multiplies with every product. Following the rules for `side2`: a difference of two inputs is `Expansion<2>`; $(q - p_i) + (q - p_a)$ is `<4>`; one product $d_x s_x$ is $2 \cdot 2 \cdot 4 = 16$; f_{ia} , three such products and a weight difference, is `<50>`; one product of two f values is $2 \cdot 50 \cdot 50 = 5000$; the numerator is `Expansion<10000>`, 80 kB on the stack. `side3` comes to about $3 \cdot 10^6$ components. These are worst cases of the formula; the real length of a value is set by how many bits it has, which is far less, and known only once it is computed.

5.3 `DynamicExpansion`: the length at runtime

`DynamicExpansion` (`dynamic_expansion.h`) keeps its first 32 components in an array inside the object and moves *all* of them to a `std::vector` when a 33rd arrives. Most exact evaluations are short (the filter only sends inputs near a degeneracy), so they never touch the heap. Because nothing points into the inline array, the compiler-written copy and move are correct. One caveat of the default move: a heap-stored expansion that has been moved from keeps its count while its vector is (in practice) empty, so it must be assigned to before it is read again, as with any moved-from object. Its free functions mirror `expansion.h`'s, so a formula reads the same with either type, with two differences: `product` needs no `fit`, and `a * b` between two expansions is `compress(product(a, b))`. The reason is cost: a product costs at least the product of its factors' lengths, so without compression a degree-6 formula gets slower at every level; one compress pass keeps each level short. Sums are not compressed: a sum is at most as long as its inputs together.

	Expansion<N>	DynamicExpansion
storage	<code>std::array<double, N></code>	32 inline, then <code>std::vector</code>
length bound	in the type, checked at compile time	none
used by	<code>orient2d, orient3d</code>	<code>side1</code> to <code>side4</code>
<code>a * b</code>	<code>product</code>	<code>compress(product)</code>

6 The filter: Bounded

6.1 The idea

A **Bounded** (`bounded.h`) is a double that knows how wrong it might be: a value and a bound, with one promise,

$$|\text{value} - \text{exact}| \leq \text{error},$$

where *exact* is what the same formula would give in exact arithmetic. Inputs are exact (error 0). Each operation computes its value in plain double, as fast as ever, and a new bound alongside: the operands' errors carried through, plus the operation's own rounding. At the end, if $|\text{value}| > \text{error}$, the exact value cannot be zero or have the other sign, so the sign of the value is proven.

6.2 The rules, and why they are sound

Let $\hat{a}, \hat{b}$ be the computed values and a, b the exact ones, with $|\hat{a} - a| \leq \epsilon_a$ and $|\hat{b} - b| \leq \epsilon_b$. The operation rounds once, which adds at most u times the result.

Sum or difference. $\hat{a} \pm \hat{b}$ differs from $a \pm b$ by at most $\epsilon_a + \epsilon_b$, and rounding adds at most $u|\text{fl}(\hat{a} \pm \hat{b})|$:

$$\epsilon_{a \pm b} = \epsilon_a + \epsilon_b + u|\text{value}|.$$

Product. $\hat{a}\hat{b} - ab = \hat{a}(\hat{b} - b) + \hat{b}(\hat{a} - a) - (\hat{a} - a)(\hat{b} - b)$, so

$$\epsilon_{ab} = |\hat{a}| \epsilon_b + |\hat{b}| \epsilon_a + \epsilon_a \epsilon_b + u|\text{value}|.$$

Negation. Exact: the bound is unchanged.

By induction over the operations of a formula, the promise holds for the result whenever it holds for the inputs, which it does (error 0). So the filter is *sound by construction*, for every formula at once: nothing has to be derived per predicate.

Two details turn “nearly safe” into “safe”:

- The bound is itself computed in double, and those few operations can round down. Each new bound is multiplied by `kInflate` = $1 + 2^{-48}$, which absorbs the rounding of about thirty operations ($2^{-48} = 32u$), where a product's bound has seven. It also covers using $u|\text{value}|$ where the rule needs u times the unrounded result.
- $u|\text{value}|$ is a *relative* rounding bound, true only outside the subnormals, where rounding errors become absolute: the same range the exact path needs (§2).

And two cases fail safe by themselves: a bound that overflows to infinity, or a NaN, makes $|\text{value}| > \text{error}$ false, so the filter says “don't know”. **The filter can be unsure; it cannot be wrong.**

6.3 Three answers

`certain_sign` returns a `std::optional<int>`:

condition	answer
$ \text{value} > \text{error}$	the sign of value: proven
$\text{value} = 0$ and $\text{error} = 0$	0: the exact value is zero and was never rounded
otherwise	nothing: zero is within the bound, “don’t know”

The second line is real: $x - x$ of the same input, or a product with an exact zero. An `optional` rather than an `int`, because 0 is a valid answer and must not be confused with “don’t know”.

Example. $e = ab - cd$ with exact inputs and $ab \approx cd \approx 1$. The products have bounds $u|fl(ab)|$ and $u|fl(cd)|$; the difference has bound about $u(|ab| + |cd| + |e|) \approx 2u \approx 2.2 \cdot 10^{-16}$. If $|e| \approx 10^{-12}$ the sign is proven; if $|e| \approx 10^{-17}$, the bound covers zero and the exact path decides.

6.4 Why a dynamic filter

Shewchuk’s predicates, and those generated by Meyer and Pion’s FPG, use *semi-static* filters: the error bound is a constant derived in advance for each predicate, multiplied at runtime by a magnitude computed from the input (Shewchuk calls it a “runtime error estimate”; for `orient2d` it is a sum of absolute values of the two products). That costs a few operations per call, but every new formula needs its constant derived, and a wrong constant gives wrong answers silently. `Bounded` carries a bound through every operation instead. Measured, it costs 3 to 13 times a plain double evaluation, against 180 to 1900 times for the exact path (§9), and it needs no derivation: any formula written as a template gets a filter for free. A semi-static filter can be put in front of it later, for the hottest predicate, if a profile asks for one.

6.5 Why the formulas are translated

The bound is proportional to the magnitudes of the intermediate values. Writing a formula in *differences of inputs* – `orient2d` in $a - c$ and $b - c$, the side predicates’ f_{ia} as $(p_a - p_i) \cdot ((q - p_i) + (q - p_a)) + (w_a - w_i)$ – makes each difference round by at most half an ulp of itself, so the bound scales with the size of the configuration, not with its distance from the origin. Three points near $(10^6, 10^6)$ a millimetre apart then get a bound on the scale of their own tiny area, and the filter can still decide them. The exact path gains too: each difference is at most two components.

7 How a predicate puts them together

7.1 One formula, several number types

Each predicate’s formula is written once, as a template over the number type `T`, and run with up to two types:

T	what the same text computes
<code>double</code>	the plain formula: fast and rounded (used only by the tests, to show it fails)
<code>Bounded</code>	the filter: the value and a proven bound on its error
<code>Expansion<1></code>	the exact value, sizes worked out by the compiler (<code>orient2d</code> , <code>orient3d</code>)
<code>DynamicExpansion</code>	the exact value, sized at runtime (<code>side1</code> to <code>side4</code>)

Every input is converted to T *before* it is used: $T(\text{pa.x}) - T(\text{pi.x})$, never $T(\text{pa.x} - \text{pi.x})$. The second rounds in double first; the exact path then loses exactness, and worse, `Bounded(v)` claims error 0 for a number that is already rounded, so the filter could certify a wrong sign.

7.2 Ratios: two signs

`side2` to `side4` are ratios N/D , and the sign of a ratio is the product of the signs: nothing is ever divided. The filter’s answer is trusted in two cases only:

- both sign N and sign D are certain: the answer is their product;
- N is certainly 0: the answer is 0 whatever D is, since $D \neq 0$ is a precondition.

A certain N with an uncertain D is not enough. Otherwise both are evaluated exactly, the exact path asserts sign $D \neq 0$ (the vertex exists), and returns sign $N \cdot \text{sign } D$.

7.3 Counters

Each predicate has a `PredicateCounts` per thread (`side1_counts()` and so on), with `filtered` and `exact`, and from step 7 `perturbed`. Thread-local, so counting needs no lock; integers, so totals are exact and deterministic. The app’s robustness window is meant to show them; for now it shows a placeholder. The tests check them: random inputs must be decided by the filter (at least 95% to 99%), near-ties must reach the exact path, and with perturbation, every resolved tie is counted once.

8 Simulation of Simplicity

8.1 Why ties need more than an exact zero

After the exact path, a predicate’s answer is right, including 0 on a tie. But a clipper cannot act on 0: a vertex exactly on the bisector is neither kept nor cut, a Voronoi vertex where five cells meet has more planes than a simple vertex, and every algorithm would need special cases for each kind of degeneracy. Worse, special cases decided separately can disagree with each other, and disagreement is exactly the crack the exact arithmetic was meant to remove. On a cubic lattice of seeds, *every* Voronoi vertex is such a tie.

Simulation of Simplicity (Edelsbrunner and Mücke, 1990), in the spirit of how Geogram’s PCK handles the same predicates (Lévy, 2016), removes the ties of the side predicates: it answers every question about an imaginary input, infinitesimally close to the real one, on which no side predicate about a well-defined vertex is ever 0. It does not remove every degeneracy: the domain is not perturbed (§8.5), and a vertex whose defining equations have no unique solution stays undefined.

8.2 The perturbation

Imagine every seed’s weight increased by an infinitesimal:

$$w_a \mapsto w_a + \varepsilon_a, \quad \varepsilon_0 \gg \varepsilon_1 \gg \varepsilon_2 \gg \cdots > 0,$$

where the index is the seed’s index in the point set. Every ε is infinitely small, and each is infinitely larger than the next. Concretely one can think of $\varepsilon_a = \delta^{a+1}$ for a δ that tends to 0: then any linear combination $F_0 + \sum_a F_a \varepsilon_a$ has, for small enough δ , the sign of F_0 if $F_0 \neq 0$, and otherwise the sign of the first nonzero F_a in increasing index order. That rule is the whole of the perturbation’s arithmetic.

Increasing w_a lowers $\pi_a(x) = \|x - p_a\|^2 - w_a$, so every $f_{ia} = \pi_i - \pi_a$ shifts by a constant:

$$f_{ia} \mapsto f_{ia} + (\varepsilon_a - \varepsilon_i).$$

A seed with a smaller index has the infinitely larger extra weight. At a *fixed* point, such as **side1**'s domain corner, that settles it: the smaller index wins the tie. For **side2** to **side4** the vertex itself moves when the weights move, and a third seed's ε can decide (the worked example below has one); there the rule is the coefficient rule of §8.4, not “the smaller of i and m ”.

8.3 Why perturb the weights

Weights shift; coordinates bend. A tie in a side predicate is always a tie between two power distances, and a weight is exactly what moves a power distance: $w_a + \varepsilon_a$ lowers π_a by ε_a at every point, so f_{ia} shifts by the constant $\varepsilon_a - \varepsilon_i$. Perturbing a coordinate instead, $p_a + \delta_a$, gives

$$\|x - p_a - \delta_a\|^2 = \|x - p_a\|^2 - 2(x - p_a) \cdot \delta_a + \|\delta_a\|^2 :$$

a change that depends on x , second-order terms, and three infinitesimals per seed. Through a determinant of degree 6 this becomes a polynomial in many infinitesimals, with products and powers of them; its sign needs an order on all those monomials and a long sequence of minors, worked out for each predicate, as in Edelsbrunner and Mücke's original scheme. The first-order terms can even all vanish: if $q = p_i = p_m$ and $w_i = w_m$, **side1**'s perturbed value is $\|\delta_i\|^2 - \|\delta_m\|^2$, decided only at second order.

Constant shifts keep it linear. Take **side3** as the example; **side2** is the same one size smaller. Its numerator is a 3×3 determinant whose rows are the values of three f functions at the triangle's corners q_0, q_1, q_2 :

$$N = \det(M; J; K) = \begin{vmatrix} m_0 & m_1 & m_2 \\ j_0 & j_1 & j_2 \\ k_0 & k_1 & k_2 \end{vmatrix}, \quad D = \det(\mathbf{1}; J; K) = \begin{vmatrix} 1 & 1 & 1 \\ j_0 & j_1 & j_2 \\ k_0 & k_1 & k_2 \end{vmatrix}.$$

The symbols:

- $M = (m_0, m_1, m_2)$ is a row of three numbers, $m_r = f_{im}(q_r)$; likewise $J = (j_0, j_1, j_2)$ with $j_r = f_{ij}(q_r)$, and $K = (k_0, k_1, k_2)$ with $k_r = f_{ik}(q_r)$;
- $\mathbf{1} = (1, 1, 1)$ is the row of ones;
- $\det(X; Y; Z)$ is the 3×3 determinant with rows X, Y, Z , top to bottom;
- $a = \varepsilon_m - \varepsilon_i$, $b = \varepsilon_j - \varepsilon_i$, $c = \varepsilon_k - \varepsilon_i$ are single numbers (infinitesimal), not rows.

Under the perturbation every value of f_{im} grows by the same number a , whichever corner it is taken at, so the row M becomes $M + a\mathbf{1} = (m_0 + a, m_1 + a, m_2 + a)$; likewise J becomes $J + b\mathbf{1}$ and K becomes $K + c\mathbf{1}$. Two facts about determinants then do all the work:

- a determinant is *linear in each row separately*: with the other rows fixed, for any number t ,

$$\det(X + tY; \dots) = \det(X; \dots) + t \det(Y; \dots);$$

for example $\det(M + a\mathbf{1}; J; K) = \det(M; J; K) + a \det(\mathbf{1}; J; K)$;

- a determinant with *two equal rows* is 0.

Splitting all three rows this way gives eight terms. Every term that keeps $\mathbf{1}$ in two rows or more vanishes, for example $ab \det(\mathbf{1}; \mathbf{1}; K) = 0$, and four are left:

$$\det(M + a\mathbf{1}; J + b\mathbf{1}; K + c\mathbf{1}) = \det(M; J; K) + a \det(\mathbf{1}; J; K) + b \det(M; \mathbf{1}; K) + c \det(M; J; \mathbf{1}).$$

No term has a product of two ε s: the perturbed numerator is linear in them. The same splitting of $D = \det(\mathbf{1}; J + b\mathbf{1}; K + c\mathbf{1})$ leaves only $\det(\mathbf{1}; J; K)$, since its first row is already $\mathbf{1}$. (**side4** is built differently but behaves the same: its weights all sit in one *column* of $\det_4$, the column of the c_a , each c_a gains $\varepsilon_i - \varepsilon_a$, and a determinant is linear in each column too; $\det_3$ has no weights at all.) So:

- the **denominator never changes**: for **side2** and **side3** its first row is already the ones, and the other shifts produce determinants with two rows of ones; **side4**'s $\det_3$ contains no weights at all. So the vertex's equations have a unique solution in the perturbed input exactly when they do in the real one. (Whether that solution lies inside the feature, or inside the cell, is a different question, answered by other predicates, and can change.)
- the **numerator becomes linear** in the ε : no products of them, so its perturbed sign is the first nonzero of at most five coefficients;
- the **coefficient of ε_m is $\pm D$** , the vertex's own denominator, which is nonzero by the predicate's precondition (a unique vertex). So the chain of coefficients always ends, and no perturbed side predicate on a well-defined vertex is ever 0.

What weights do not remove. Because denominators do not change, degeneracies that do not involve power distances stay: an edge parallel to a bisector (**side2**), two bisectors meeting a triangle's plane in a line (**side3**), four coplanar seeds (**side4**). Perturbing coordinates would remove those; here they are preconditions instead, $D \neq 0$, asserted on the exact path. That is enough because of the rule a clipper follows (the ones of DESIGN.tex, sections 7.2–7.3, classify every vertex with a side predicate): a new vertex is made only on an edge between a kept vertex and a removed one, so only where the edge's endpoints are *strictly* on opposite sides of the new bisector – strictly, because no perturbed side predicate is 0. The edge lies on a line fixed by the domain feature and by earlier bisectors. A weight shift moves each bisector parallel to itself, so that line keeps its direction, and along it the perturbed f_{im} is affine with the same slope as the real one. Opposite signs at the two endpoints mean that slope is not zero: the line crosses the bisector in exactly one point, the new vertex's system has a unique solution, and its $D \neq 0$. So every vertex the clipper creates meets the precondition. This argues about code that is not written yet; the assert on D is what will check it.

Other predicates. `orient2d` and `orient3d` contain no weights, so the weight perturbation cannot break their ties. `in_power_sphere` does contain them, all in one column, so its perturbation is linear too, but its coefficients are $\pm \text{orient3d}$ of four of its five points, which can themselves be 0. None of them has to decide a tie in the current plan: clipping uses only the side predicates, and `orient3d` serves validation, where 0 is an answer. If one ever must (the optional Delaunay of DESIGN.tex, section 7.1.3), the coordinates have to be perturbed as well, by infinitesimals infinitely smaller than every weight's. The side predicates are then unchanged – their first nonzero term is always a weight term, at worst $\pm D$ for ε_m , which outweighs every coordinate term – and all predicates still answer about one and the same imaginary input.

8.4 The coefficients, predicate by predicate

Write $a = \varepsilon_m - \varepsilon_i$, $b = \varepsilon_j - \varepsilon_i$, $c = \varepsilon_k - \varepsilon_i$, and keep the notation of the predicates document: m_r, j_r, k_r are f_{im}, f_{ij}, f_{ik} at corner q_r , and M, J, K the rows of them.

side1 $f_{im}(q) + \varepsilon_m - \varepsilon_i$. At a tie the sign is that of $\varepsilon_m - \varepsilon_i$: **-1 if i has the smaller index, +1 if m has.**

side2 $(m_0 + a)(j_1 + b) - (m_1 + a)(j_0 + b)$; the ab terms cancel:

$$N + \varepsilon_m D + \varepsilon_j(m_0 - m_1) - \varepsilon_i(D + m_0 - m_1), \quad D = j_1 - j_0.$$

side3 by the identity above,

$$N + a D + b \det(M; \mathbf{1}; K) + c \det(M; J; \mathbf{1}),$$

so the coefficients are ε_m : D , ε_j : $\det(M; \mathbf{1}; K)$, ε_k : $\det(M; J; \mathbf{1})$, and ε_i : minus their sum.

side4 each $c_a = \|d_a\|^2 + w_i - w_a$ gains $\varepsilon_i - \varepsilon_a$. $\det_4$ is linear in its column of c s and $\det_3$ holds none, so with C_a the cofactor of c_a (signs $-, +, -, +$ down the column; $C_m = \det_3 = D$) the numerator $-\det_4$ becomes

$$-\det_4 + \sum_{a \in \{j,k,l,m\}} \varepsilon_a C_a - \varepsilon_i \sum_a C_a.$$

In every case the perturbed answer is

$$\text{sign } D \cdot \text{sign}(\text{the first nonzero coefficient, seeds taken by increasing index}),$$

used only when the exact N is 0. The coefficients must be exact values (computed as a `DynamicExpansion`): a coefficient that is truly zero must be seen as zero, or the chain stops at rounding noise.

Example (side2 at a tie). The edge from $q_0 = (-1, 0, 0)$ to $q_1 = (3, 0, 0)$ crosses the bisector of $p_i = (0, 0, 0)$ and $p_j = (2, 0, 0)$ at $x = (1, 0, 0)$, and $p_m = (1, 1, 0)$ is exactly as far from x as p_i (all weights 0). The f values are $m_0 = -4$, $m_1 = 4$, $j_0 = -8$, $j_1 = 8$, so $N = m_0 j_1 - m_1 j_0 = 0$ and $D = 16$. The coefficients: $\varepsilon_m: 16$, $\varepsilon_j: m_0 - m_1 = -8$, $\varepsilon_i: -(16 - 8) = -8$.

- i has the smallest index: the first coefficient is -8 , the answer -1 . Seed i 's larger extra weight puts x on its side.
- m has the smallest index: $+16$, the answer $+1$, for the same reason the other way.
- j has the smallest index: -8 , the answer -1 . Seed j 's extra weight pushes the bisector of i and j towards i , so the crossing moves to $x = (1 - \eta, 0, 0)$, where $\pi_i \approx 1 - 2\eta$ is below $\pi_m \approx 1$: on i 's side. The coefficient says the same without the picture.

8.5 What it buys, and what to keep in mind

- **Consistency.** Every predicate answers about the same perturbed input, which is a real weighted point set, so all decisions agree and the output is a valid diagram: that of the perturbed seeds.
- **No special cases in the clipping.** A clipper never meets a vertex exactly on the bisector it clips by, and a Voronoi vertex inside a 3D cell always has exactly three bisector planes.
- **Honest geometry.** The perturbation is infinitesimal, so measures do not change: a cube corner where eight cells meet becomes several vertices at the same position, joined by zero-length edges, which integrate to nothing.
- The **domain is not perturbed**; degenerate domain pieces are removed when the domain is built.
- The tie-break follows the **seed index**: reordering the seeds (Morton order on or off) gives different, equally valid combinatorics on degenerate input, with the same measures.

8.6 In the code (step 7)

The design chosen: new functions `side1_sos` to `side4_sos` in `side.h`, taking a `uint32_t` index before each seed, which never return 0; `side1` to `side4` stay as they are, exact with 0 on a tie, for the tests and for a “no perturbation” switch. Each `sideK_sos` calls `sideK`; if that returns 0, it adds one to the new `PredicateCounts::perturbed`, computes the coefficients exactly, and returns $\text{sign } D$ times the sign of the first nonzero one by increasing index.

8.7 How the perturbation is tested

`tests/sos_support.h` holds an oracle that follows the definition, not the formulas: it solves the vertex exactly with every weight carrying its ε symbolically. The system’s matrix holds no weights, so only its right side carries ε s, and the solution and then $\pi_i(x) - \pi_m(x)$ are linear in them: $F_0 + \sum_a F_a \varepsilon_a$, with the sign rule above. That oracle is itself checked by doing what the ε s stand for: adding a real $\delta^{\text{rank}+1}$, $\delta = 2^{-100}$, to each weight and solving that input exactly in rationals. The suites then require, on a lattice full of ties (moved by 2^{20} , scaled, and with ids relabelled): the perturbed predicate is never 0; it equals the exact one wherever that is not 0; it equals the perturbed oracle at ties; the counter counts exactly the ties; and the vertex symmetries (swapping i and j , reordering corners) still hold at ties.

One lesson from validating those tests: on a small integer lattice every coefficient is a small integer, so coefficients computed in plain doubles happen to be exact and a broken implementation passes. The suites therefore also scale the lattice by a 24-bit factor (weights by its square, still exact) and, unweighted, by a 40-bit one, where doubles round.

9 Measurements

The numbers this document quotes, with how they were obtained. They come from the program in `tests/measure_side.cpp`. Once `side1` to `side4` exist, the build makes it as the executable `rvd-measure-side` in the build directory; it checks nothing and is not run by `cctest`. Run it again after a change to the kernel and update this section.

Setup. 5 October 2026; Intel Core Ultra 7 265H; GCC 16.2.1; the project’s Release build (`-O3 -DNDEBUG -ffp-contract=off`); the user’s `side.h` as of that date.

Rates. On the input families of `tests/side_support.h`, with the side suites’ own seeds: random inputs (coordinates with exponents in $[-20, 20]$, half of them weighted), a lattice of small integers, and lattice ties moved by one ulp. “Doubles wrong” constructs the vertex in doubles by Gaussian elimination and compares the two power distances, as a floating-point Voronoi code would. These are deterministic: the same on every run.

	filter decides (random inputs)	lattice ties	nudged ties reaching exact	doubles wrong on nudged ties
<code>side1</code>	100.00%	4.54%	88.1%	76.3%
<code>side2</code>	100.00%	3.26%	91.6%	71.8%
<code>side3</code>	98.19%	2.77%	91.6%	67.7%
<code>side4</code>	99.99%	3.98%	93.6%	65.2%

Timings. Nanoseconds per call on 20000 random inputs, best of 5 runs, over four runs of the program (they agreed to about 10%). Each formula is evaluated as `double`, as `Bounded` and as `DynamicExpansion`; “predicate” is the whole call, filter first and exact when needed. These are microbenchmarks: read the ratios, not the absolute times, and expect them to move with the machine and the compiler.

	double	Bounded	DynamicExpansion	predicate
side1	2.7	8.2 ($\times 3$)	495 ($\times 180$)	8.6
side2	4.7	37.5 ($\times 8$)	3200 ($\times 690$)	46
side3	14	81 ($\times 6$)	11600 ($\times 850$)	296
side4	4.6	59 ($\times 13$)	9000 ($\times 1950$)	72
orient3d	1.4	5.8 ($\times 4$)	–	11.5

The whole predicate costs about what the filter does, except for **side3**, where the 1.8% of random calls that go exact, at about $11.6 \mu\text{s}$ each, are most of the time. The plain **double** times are only a reference: that evaluation is never used for an answer.

10 Where everything lives

idea	code	tested by
EFTs	<code>eft.h</code>	<code>test_eft</code>
<code>Expansion<N></code> , <code>grow</code> , <code>sum</code> , <code>scale</code> , ...	<code>expansion.h</code>	<code>test_expansion</code> , <code>test_product</code>
<code>fast_sum</code> (and Theorem 13)	both expansion headers	<code>test_fast_sum</code> , <code>test_dynamic_expansion</code>
<code>linear_sum</code>	both expansion headers	<code>test_linear_sum</code> , <code>test_dynamic_linear_sum</code>
<code>DynamicExpansion</code> , <code>compress</code> the filter	<code>dynamic_expansion.h</code> <code>bounded.h</code>	<code>test_dynamic_expansion</code> <code>test_bounded</code> , <code>test_orient2d_filter</code>
counters	<code>predicate_counts.h</code>	every predicate suite
<code>orient2d</code> , <code>orient3d</code>	<code>orient2d.h</code> , <code>orient3d.h</code>	<code>test_orient2d</code> , <code>test_orient3d</code>
side predicates	<code>side.h</code>	<code>test_side1</code> to <code>test_side4</code>
perturbation	<code>side.h</code> (step 7)	<code>test_side1_sos</code> to <code>test_side4_sos</code>

Every kernel suite includes checks against exact rationals (GMP, `tests/exact.h`), and the suites compile with asserts on, so the precondition of `fast_two_sum` and the ordering and capacity checks of `append` run on every input the tests make; several bugs were found only by those asserts.

11 Lessons worth keeping

- A predicate wants a sign, and the sign of a polynomial in doubles can be computed exactly. Never decide from a constructed coordinate.
- Convert to the number type before subtracting; a rounded input is invisible to the filter.
- The filter can be unsure but never wrong; a certain zero is a real answer.
- The sign of an expansion is its largest component's; its value is not.
- Fast-Expansion-Sum's output is only nonoverlapping, not strongly: do not chain it. Use Linear-Expansion-Sum, which needs only nonoverlap.

- Compress is not idempotent in length; do not test for that.
- Exactly collinear points with exact coordinate differences give an exact 0 even in doubles (both products round identically); doubles fail *near* the line.
- Range is about every intermediate product, not only the inputs: keep inputs zero or of moderate size, so that no product falls below 2^{-969} . A squared subnormal, or a product of two normal numbers near 2^{-600} , underflows and exactness is gone.
- In tests, small integer inputs make doubles exact by luck. Scale them to make rounding show.
- Under perturbation, denominators never change and numerators are linear in the ε s; the coefficient of ε_m is $\pm D$, so ties on a well-defined vertex always break. Who wins is the coefficient rule, not “the smaller index”, except at a fixed point.
- Quote measured numbers with how they were measured; `rvd-measure-side` reproduces the ones here.

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